

1 Definition and Purpose of Gates

Quantum computers are build from quantum circuits, which consists of wires and quantum gates. Quantum gates are designed to manipulate qubits to perform some operations, and achieve phenomena like superposition and entanglement. They can be represented mathematically by a unitary matrix, and thus their operations can be reversed by applying its inverse (which is the conjugate transpose)

1.1 Frequently used quantum gates

Note: Before learning about the different gates, there are some points to keep in mind:

- Quantum gates are represented by certain symbols (like H for Hadamard gate for example). The symbol followed by the qubit state signifies an operation on that system. $(H|0\rangle)$ represents the application of Hadamard gate on the qubit $|0\rangle$.
- Quantum gates are linear operators, which is best explained with the help of the following equation :

$$H(\alpha|0\rangle + \beta|1\rangle) = \alpha H|0\rangle + \beta H|1\rangle$$

1.1.1 Single qubit gates

- **Hadamard (H):** $\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$
 - Used to create superposition.
 - $|0\rangle \rightarrow \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$
 - $|1\rangle \rightarrow \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$
 - Bloch sphere effect: Rotation by π about the axis $\frac{1}{\sqrt{2}}(\hat{x} + \hat{z})$ (swaps the x - and z -axes and negates the y -axis).
- **Pauli-X (X):** $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
 - Acts like a quantum-NOT gate.
 - $|0\rangle \rightarrow |1\rangle$
 - $|1\rangle \rightarrow |0\rangle$
 - Bloch sphere effect: Rotation by π about the x -axis.
- **Pauli-Y (Y):** $\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$
 - $|0\rangle \rightarrow i|1\rangle$
 - $|1\rangle \rightarrow -i|0\rangle$
 - Bloch sphere effect: Rotation by π about the y -axis.
- **Pauli-Z (Z):** $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
 - $|0\rangle \rightarrow |0\rangle$
 - $|1\rangle \rightarrow -|1\rangle$

- Bloch sphere effect: Rotation by π about the z -axis.

- **Phase (S):** $\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$

- $|0\rangle \rightarrow |0\rangle$
- $|1\rangle \rightarrow i|1\rangle$
- Bloch sphere effect: Rotation by $\pi/2$ about the z -axis.

- $\pi/8$ (T): $\begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$

- $|0\rangle \rightarrow |0\rangle$
- $|1\rangle \rightarrow e^{i\pi/4}|1\rangle$
- Bloch sphere effect: Rotation by $\pi/4$ about the z -axis.
- Since $e^{i\pi/4}$ is the square root of i , $\pi/8$ gate (T gate) is the square root of Phase gate (S gate), which itself is the square root of Pauli-Z gate

- $R_x(\theta)$: $\begin{bmatrix} \cos \frac{\theta}{2} & -i \sin \frac{\theta}{2} \\ -i \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{bmatrix}$

- Rotation by angle θ about the x -axis.
- $|0\rangle \rightarrow \cos \frac{\theta}{2}|0\rangle - i \sin \frac{\theta}{2}|1\rangle$
- $|1\rangle \rightarrow -i \sin \frac{\theta}{2}|0\rangle + \cos \frac{\theta}{2}|1\rangle$
- Bloch sphere effect: Rotation by θ about the x -axis.

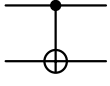
- $R_y(\theta)$: $\begin{bmatrix} \cos \frac{\theta}{2} & -\sin \frac{\theta}{2} \\ \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{bmatrix}$

- Rotation by angle θ about the y -axis.
- $|0\rangle \rightarrow \cos \frac{\theta}{2}|0\rangle + \sin \frac{\theta}{2}|1\rangle$
- $|1\rangle \rightarrow -\sin \frac{\theta}{2}|0\rangle + \cos \frac{\theta}{2}|1\rangle$
- Bloch sphere effect: Rotation by θ about the y -axis.

- $R_z(\theta)$: $\begin{bmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{bmatrix}$

- Rotation by angle θ about the z -axis.
- $|0\rangle \rightarrow e^{-i\theta/2}|0\rangle$
- $|1\rangle \rightarrow e^{i\theta/2}|1\rangle$
- Bloch sphere effect: Rotation by θ about the z -axis.

1.1.2 Multi qubit gates

• **Controlled-NOT (CNOT):** 
$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

– Flips the target qubit if and only if the control qubit is $|1\rangle$.

* $|01\rangle \rightarrow |01\rangle$

* $|11\rangle \rightarrow |10\rangle$

• **SWAP:** 
$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

– Exchanges the states of the two qubits.

* $|01\rangle \rightarrow |10\rangle$

• **Controlled-Z:** 
$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

– Applies a phase flip (-1) to the target qubit if and only if the control qubit is $|1\rangle$.

* $|01\rangle \rightarrow |01\rangle$

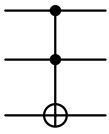
* $|11\rangle \rightarrow -|11\rangle$

• **Controlled-Phase:** 
$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & i \end{bmatrix}$$

– Applies a phase of i to the target qubit if and only if the control qubit is $|1\rangle$.

* $|01\rangle \rightarrow |01\rangle$

* $|11\rangle \rightarrow i|11\rangle$

• **Toffoli (CCNOT):** 
$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

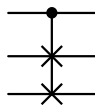
– Flips the target qubit if and only if both control qubits are $|1\rangle$.

* $|001\rangle \rightarrow |001\rangle$

* $|101\rangle \rightarrow |101\rangle$

* $|110\rangle \rightarrow |111\rangle$

- **Fredkin (Controlled-SWAP):**



$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

- Swaps the two target qubits if and only if the control qubit is $|1\rangle$.

$$* |100\rangle \rightarrow |100\rangle$$

$$* |101\rangle \rightarrow |110\rangle$$

1.2 Quantum circuits

Quantum circuits are sequence of quantum logic gates, measurements, and initializations that are applied to a system of qubits. The following are some symbols used in quantum circuits :

- **Measurement:**

- Projection onto $|0\rangle$ and $|1\rangle$.

- **Qubit wire:**

- Wire carrying a single qubit (time goes left to right).

- **Classical bit wire:**

- Wire carrying a single classical bit.

- **n qubits wire:**

- Wire carrying n qubits.

This is an example of a quantum circuit operating on 4 qubits that are initialized to $|0\rangle$. Try to map out the effect of the circuit on the system on your own.

